House Price Prediction – Project Summary

1. Overview

The goal of this project was to analyze housing data and build machine learning models to predict house prices based on property features such as area, bedrooms, bathrooms, parking, amenities, and furnishing status.

2. Data Preparation and Exploration

The dataset contained 545 records and was found to be clean, with no missing values or duplicate entries. Categorical variables were converted into numerical form using encoding techniques to make them suitable for machine learning models.

3. Key Findings

The analysis showed that area was the most influential factor affecting house prices. Other important features included bathrooms, parking availability, air conditioning, and preferred area. In general, houses with larger areas and better amenities tended to have higher prices.

4. Model Performance

Two models were trained and evaluated: Linear Regression and Random Forest Regressor. Linear Regression achieved the better performance, with a higher R^2 score and lower prediction errors. This suggests that the relationship between the selected features and house prices is largely linear.

5. Conclusion and Recommendation

The project demonstrates that house prices can be predicted reasonably well using property characteristics and amenities. Based on the findings, real estate businesses should focus on highlighting larger property sizes, premium amenities, and desirable locations, as these factors contribute significantly to higher property values.